Bias toward the assisting algorithm in polysomnography reference standards built from edited automatic pre-scoring

Quoc Viet Nguyen[1,2,*], Duc Hoai Anh Nguyen[3], Phuc Tai Duong[3], Dzung Van Dinh[4], Nam S. Vo[1,5] and Zengchang Qin[1,2,6]

[1]College of Engineering and Computer Science, VinUniversity, Hanoi, Vietnam

[2]GREEN-X, VinUniversity, Hanoi, Vietnam

[3]Hanoi University of Science and Technology, Hanoi, Vietnam

[4]VNU Information Technology Institute, Hanoi, Vietnam

[5]VinUni Big Data Research Institute, VinUniversity, Hanoi, Vietnam

[6]Center for AI Research, VinUniversity, Hanoi, Vietnam

[*]Author to whom any correspondence should be addressed

Article type: Research Article. **Journal section:** Sleep Technology & AI.

Authors' e-mail addresses and ORCID iDs. Quoc Viet Nguyen (corresponding author): viet.nq@vinuni.edu.vn, ORCID 0009-0000-8716-9148. Duc Hoai Anh Nguyen: ORCID 0009-0004-5679-1582. Phuc Tai Duong: ORCID 0009-0002-0650-0332. Dzung Van Dinh: ORCID 0000-0003-2333-5623. Nam S. Vo: ORCID 0000-0002-5454-9176. Zengchang Qin: ORCID 0000-0002-8084-6721.

Data availability statement. The PSG-IPA dataset analyzed in this study is openly available on PhysioNet at https://doi.org/10.13026/ESX0-NW71. The analysis plans are provided as Supplementary File S1. The analysis code with its synthetic tests, the aggregate reports and the coding of the FDA summaries are openly available at https://github.com/quocvietnguyen75/psg-assisted-scoring-bias.

Funding statement. This research received no specific grant from any funding agency in the public, commercial or not-for-profit sectors.

Conflict of interest disclosure. The authors declare no conflicts of interest.

Ethics approval statement. This is a secondary analysis of open, de-identified data. The original data collection was approved by the Medisch Ethische Toetsingscommissie Zuidwest Holland (MTEC-19-065), as reported by the dataset's creators.

Patient consent statement. Not applicable to this secondary analysis; the ethics committee of the original study waived explicit informed consent.

Permission to reproduce material from other sources. Not applicable.

Clinical trial registration. Not applicable.

Author contributions. Quoc Viet Nguyen: conceptualization, investigation (literature search), methodology, software and formal analysis (with AI assistance), data curation, validation (adjudication of the FDA coding), resources (computing), project administration, writing (original draft, with AI assistance), writing (review and editing). Duc Hoai Anh Nguyen: investigation (literature review), validation (FDA coding; re-implementation, without access to the code, of the anchoring measure and the staging inflation measures; clean re-run of the pipeline), writing (review and editing). Phuc Tai Duong: investigation (literature review), validation (FDA coding), writing (review and editing). Dzung Van Dinh: supervision, writing (review and editing). Nam S. Vo: supervision, writing (review and editing). Zengchang Qin: supervision, writing (review and editing).

Acknowledgements. We thank the creators of PSG-IPA for making paired manual, assisted and automatic scorings openly available.

Use of generative AI. Quoc Viet Nguyen directed this work and managed the computing resources; all authors take full responsibility for the content of this article. Claude Opus 5.5 and Claude Fable 5.1 (Anthropic; model identifiers claude-opus-5-5 and claude-fable-5-1), used through the Claude Code command-line tool from September to November 2026, drafted the analysis plans, which were fixed before their analyses were run; wrote and tested the analysis, simulation and figure code and ran it on the authors' computer; extracted and coded the public FDA 510(k) summaries; supported the literature search and the checking of reference metadata against DOI registries; drafted and revised the manuscript text from the study records; and ran three simulated pre-submission peer reviews, three fidelity checks of the text against those records and a final verification pass. No AI tool fabricated or altered research data or results. Every reported result is generated by the recorded analysis code, whose kappa and sign tests agreed with standard library implementations.

The authors checked the AI's work in four ways. First, one of two authors checked the AI's FDA coding against each summary; both then re-coded its three key fields for all 22 summaries by hand and, by their account, without consulting any code, each blind for the summaries the other had checked, and the corresponding author settled the disagreements from the summary text (one code changed; Supplementary Methods and Table S18). Second, a co-author who had not read the analysis code, but had seen the manuscript's results, re-implemented from the protocol text, with a different AI coding assistant (OpenAI Codex), the anchoring measure for all four tasks and the inflation measures for sleep staging, and reproduced all 35 per-recording values; the remaining quantities rest on the analysis code's synthetic tests. Third, the same co-author re-ran the whole pipeline from a fresh clone of the study repository on freshly downloaded data, with both stager environments rebuilt, and reproduced exactly every value then in the manuscript's generated values file and the intermediate reports to the last floating-point bit (Supplementary Methods). Fourth, the authors checked each citation against its source and reviewed and revised all AI-assisted content.

Abstract

Automatic sleep scorers are validated against expert reference standards, increasingly produced by editing an algorithm's pre-scoring. Taking the same experts' unassisted scoring as the standard, we asked whether such assisted references lie closer to the assisting algorithm than that standard. In the open PSG-IPA dataset, 12 technicians at one center scored 20 polysomnograms (five each for sleep stages, arousals, respiratory events and limb movements) manually and, at least 4 months later, by editing the pre-scoring. Under dated, unregistered analysis plans, assisted scorings were closer to the pre-scoring than the same experts' manual scorings in 19 of 20 recording-task units ($p = 4.0 \times 10^{-5}$ for these experts and recordings), most where experts disagreed. Where the consensus changed between sessions, it took the pre-scoring's label in 79% to 99% of the changed units (epochs for stages, 0.1-s bins for events). Where two other sleep stagers, which no expert saw, disagreed with the pre-scoring, the experts moved toward the pre-scoring and away from both stagers. Against the assisted reference, the pre-scoring's lead over each stager grew in every recording. A stager that had been level with the pre-scoring against the manual reference, ahead in 3 of 5 recordings, trailed it in all five. In an emulation with the earlier session's unassisted labels, an audit of 10% of 30-minute blocks estimated the staging comparison without this bias (95% half-width 0.045; 0.059 with the audit's own degrees of freedom). No public summary of the 22 U.S. Food and Drug Administration clearances since 2012 with performance data under product code OLZ (automatic PSG event detection) stated whether reference scorers worked from automatic pre-scoring by any software. In this dataset, references built by editing an algorithm's output favored that algorithm. Validations should report how their references were made and audit a random sample of blocks scored without assistance before any exposure to the algorithm's labels for them.

Keywords: anchoring; automatic sleep scoring; incorporation bias; polysomnography; reference standard; validation

1 Introduction

New automatic polysomnography (PSG) scorers are judged by how closely they agree with expert scorers [1, 2]. Because experts disagree, most of all on N1 sleep and hypopneas [3–7], careful validations compare algorithms with the consensus of several scorers [2, 8]. The reference standard is thus a product of human scoring, whose properties decide what a validation can show.

Cleared scoring software is intended to be reviewed and corrected by a clinician [9]. Workflows in which technologists edit an automatic scoring, or review only the epochs flagged as uncertain, have been described [10–12]. Such assisted scorings also enter validation. In a large validation of an arousal detector, part of the clinical reference had been produced by reviewing the detector's own output, and the detector agreed better with that part (kappa 0.600) than with the manual part (0.559) [13].

Assisted scoring raises agreement between experts. In the Somnolyzer validation, two assisted scorings agreed at kappa 0.99 against 0.76 for two manual scorings, after experts had changed about 4% of the automatic epochs [14]. Agreement rose from 78.9% to 86.5% when the same technologists edited an automatic scoring [15]. In the study behind this dataset, inter-scorer kappa rose in all four tasks [16]. These gains have been read as reduced variability or better quality [14, 16]. The arousal-detector study concluded that its algorithm "behaves as one expert more"

[13].

There is another explanation. Assisted experts may agree more because they start from the same pre-scoring and keep much of it. People anchor on suggested values [17], and clinicians follow automated advice, even when wrong [18, 19]. Correct recommendations raised sleep technologists' staging accuracy, and incorrect ones lowered it by 3 to 7 percentage points [20]. Annotators who edit a model's output agree more with it [21, 22]. Diagnostic-accuracy research calls a reference standard influenced by the test under evaluation incorporation bias, which is expected to inflate the test's measured accuracy [23–25]. If assisted references carry an analogous, behavioral form of this bias, validations against them overstate the assisting algorithm and handicap its competitors.

Separating the two explanations requires the same experts' manual and assisted scorings of the same recordings, the unedited pre-scoring, and raters who never saw it. PSG-IPA provides all three [16, 26]. Neither its creators [16] nor the planted-error experiments [20] asked whether a reference built from assisted scorings stays independent of the assisting algorithm, as a validation needs. In dated analysis plans, we asked whether assisted scorings move toward the pre-scoring, and whether references built from them favor the assisting algorithm, relative to the same experts' unassisted scoring and over algorithms the experts never saw. We also asked where the pull arises and how precisely a small unassisted audit can estimate the comparison, and reviewed how cleared scoring devices built their references.

2 Methods

2.1 Data

PSG-IPA (version 1.0.0, open access under CC BY 4.0) contains 20 clinical polysomnograms selected from the 2,801 recorded in 2019 at the Haaglanden Medisch Centrum Sleep Center, The Hague, the Netherlands [16, 26–28]. The 20 patients (12 men) had a median age of 57 years (interquartile range 51.8 to 61.5) and a median time in bed of 7.3 hours [26]. Five recordings are provided for each of four tasks: sleep staging, electroencephalographic (EEG) arousals, respiratory events and limb movements. Stages, arousals and respiratory events were scored by version 2.4 of the American Academy of Sleep Medicine (AASM) manual, and limb movements by the 2016 World Association of Sleep Medicine (WASM) standard [29]. An automatic procedure selected, for each task, the recordings at the 12.5th, 37.5th, 50th, 62.5th and 87.5th percentiles of the agreement between the automatic output and the original clinical scoring. All had been accepted for manual scoring in the center's routine workflow [16, 26].

Twelve experienced, certified sleep technicians from the center's daily scoring routine scored every recording of a task twice in the Polyman software [30]. They were blinded to patient identity and to earlier scorings, and were not told that the two sessions used the same recordings [16]. The first session was manual. In the second, at least four months later, the recording was pre-filled with the output of an automatic algorithm for that task, the *pre-fill*, and the experts reviewed it, adding, deleting or editing stages and events. We call these scorings *assisted*; the dataset calls them semi-automatic. The pre-fills came from research algorithms by the dataset's first author and colleagues: a deep-learning stager developed on this center's clinical scorings [31], an EEG arousal detector [13], and detectors of limb movements and respiratory events (Supplementary

Methods). The documentation gives no version numbers and does not say whether the algorithms were used in routine scoring (see Limitations). The dataset distributes each recording's pre-fill with the 24 scorings. All 1,025 files were verified against the published checksums. One assisted staging, SN1 by expert 4, is byte-identical to the pre-fill: the expert accepted it without edits. It is kept: the plans foresaw no exclusion. The original study was approved by a medical ethics committee, which waived explicit informed consent [16].

2.2 Analysis plan and estimand

All analyses were fixed in dated written plans before they were computed: a protocol and five amendments. The last two amendments were written after pre-submission reviews; the fifth added derived quantities only. Supplementary File S1 reproduces them verbatim, with their times and repository commits. The plans were not publicly registered. The protocol and the first amendment were committed together with their results, so their order rests on time-stamped working logs. Each analysis amendment stated its predictions before its analyses were run. Every analysis was run once; the simulation was run once more after a dated note, before any of its consequences had been computed. Every result is reported, including failed predictions (Supplementary Table S10). Supplementary Table S1 maps the names used here to the plan codes.

The truth is unknown in these recordings, so we did not ask which reference is more accurate. The standard was the same experts' unassisted scoring, the field's reference standard, and the question was whether references built from assisted scorings lie closer to the assisting algorithm than that standard does.

2.3 Agreement and reference standards

Analyses were restricted to the interval between the "lights off" and "lights on" annotations. A 30-second epoch counted when its midpoint lay in the interval, and events were clipped to it. Sleep stages (W, N1, N2, N3, R) were compared with five-class Cohen's kappa [32], dropping epochs left unscored by either scoring. Respiratory events of all types were pooled, excluding respiratory-effort-related arousals. Agreement between two event scorings was the event-level F1 score, with events matched one to one, largest overlap first (Supplementary Methods).

We built a consensus reference separately from the 12 manual and the 12 assisted scorings. A stage epoch took the label chosen by at least 7 of the 12 experts, and had no consensus otherwise. For events, a 0.1-second bin counted as an event when at least 7 of 12 scorings covered it, and contiguous bins formed consensus events. Leave-one-out consensuses of the other 11 experts required at least 6 of 11.

2.4 Primary measures and inference

Anchoring. For expert s and recording r,

$$\Delta A(s, r) = \mathrm{agr}\big(\mathrm{assisted}(s, r),\ \mathrm{prefill}(r)\big) - \mathrm{agr}\big(\mathrm{manual}(s, r),\ \mathrm{prefill}(r)\big). \tag{1}$$

Reference inflation. For each recording, Δ_{auto} is the change in the pre-fill's agreement from the manual to the assisted consensus. Δ_{human} is the same change for each expert's manual scoring,

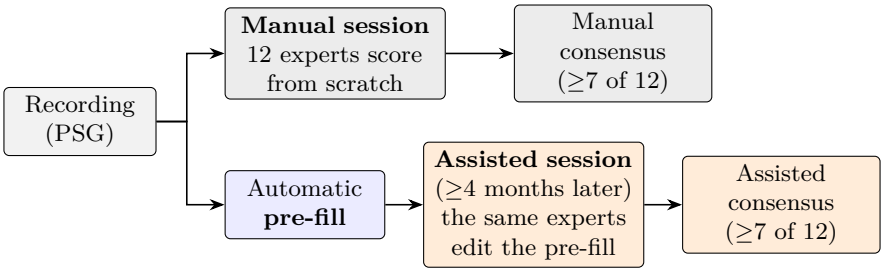


Figure 1: Design. PSG-IPA provides, for each recording, 12 manual scorings, 12 assisted scorings by the same experts, and the pre-fill that the assisted scorings started from. Five recordings are provided for each of four tasks. **Anchoring:** each expert's agreement with the pre-fill, assisted minus manual. **Inflation:** the pre-fill's agreement gain from the manual to the assisted consensus, minus the gain of each expert's manual scoring against the other 11 experts' consensus. **Independent algorithms:** two sleep stagers that no expert saw, judged against both consensuses.

judged against the other 11 experts' consensus of each session and averaged over the 12 experts. If assistance only made the reference more accurate, these raters, who had not seen the pre-fill, would gain about as much as the algorithm. The difference in differences, $\text{DiD} = \Delta_{\text{auto}} - \Delta_{\text{human}}$, would then be close to zero. Each expert's manual scoring shares its session with the manual consensus, so any difference between sessions unrelated to the pre-fill also lowers Δ_{human}. We therefore interpret the DiD as inflation relative to session-matched raters. The independent algorithms below have no session.

Inference. The units were the 20 recording-task combinations. The primary tests were exact two-sided sign tests over these units ($\alpha = 0.05$), applied to the recording-level mean of ΔA and to the DiD. Because all units share the same 12 experts, the tests support conclusions about these experts on these recordings only. Per-task means come with 95% crossed-bootstrap intervals that resample experts and recordings independently [33]. With five recordings per task they are descriptive. The many secondary sign tests and intervals are unadjusted and are read as descriptions, not separate confirmations. The pre-specified decision rule called the result positive if both sign tests were significant in the positive direction and at least one task had a mean ΔA of 0.05 or more, a threshold set by judgment.

2.5 Secondary analyses

We computed pairwise percent agreement among the 12 scorings of each session, separately where the pre-fill agreed with the manual consensus and where it disagreed. We computed events per hour of sleep for the pre-fill and both consensuses, and the *acceptance lift*: the probability that an assisted label equals the pre-fill's label, minus the same probability for the manual label. The lift was computed by the stage the pre-fill assigned and by how many of the 12 manual scorings shared the most frequent label. It was also computed on labels that most experts had rejected: units where at least 9 of the other 11 experts' manual labels differed from the pre-fill's. Selecting these units without the scored expert's own label avoids the regression to the mean of the first amendment's rule. For events, the units were whole pre-fill events, adopted when an expert's scoring overlapped them. *Side-taking* compares, among units where both consensuses exist and differ, the share where the assisted consensus equals the pre-fill with the share where the manual consensus does (equal if the sessions were exchangeable; exact sign test over the 20

units). Anchoring was also computed for apneas and hypopneas separately. The primary tests were repeated under four robustness variants, and the inflation test also with consensus events shorter than the rules' minimum durations dropped (Supplementary Methods).

Two other stagers, independent of the pre-fill and never shown to any expert, scored each sleep-staging recording with their default pretrained models: YASA 0.7.0 [34] and GSSC 0.0.9 [35], using the C4-M1 EEG (Cz-M1 in one recording) and E1-M2 electrooculogram channels (YASA also chin electromyography). The protocol named YASA, the fourth amendment added GSSC, and their settings were fixed before they were run. For each stager we report kappa against both consensuses and the pre-fill's lead over it, the pre-fill's kappa minus the stager's against the same consensus. The lead over GSSC was pre-specified; the lead over YASA was an exploratory addition to the protocol's YASA check. An assisted panel reaches a consensus on more hard epochs, which alone lowers an independent rater's measured agreement. Changes were therefore also computed on epochs where both consensuses exist, the *common support*. As a negative control, on epochs where a stager and the pre-fill disagreed, we computed each expert's movement toward the stager's label and toward the pre-fill's, the assisted minus manual share of matching labels. A general shift toward algorithm-like or more typical scoring would move experts toward the stagers too.

A two-label latent-class model calibrated to the staging data illustrated how anchoring propagates into references of different sizes and rules. Its truth is, by construction, what a large unassisted panel would converge to, and its outputs are illustrations, not estimates (Supplementary Methods).

2.6 Unassisted audit

We emulated a validation that scores every epoch with assistance and has the same experts score a random sample of 30-minute blocks without the pre-fill, drawn within each recording. We compared four estimates of an algorithm's agreement with the unassisted consensus: the assisted consensus alone; a 9-of-12 assisted consensus; the audited blocks alone; and a *difference estimator* (prediction-powered inference [36]). The difference estimator adds the mean, over audited epochs, of the unassisted minus the assisted agreement to the mean assisted agreement over all epochs. The targets were the pre-fill's and YASA's percent agreement and their gap, on the 4,857 sleep-staging epochs (87 blocks) where both algorithms are labelled and the manual consensus exists. Standard errors are exact and design-based [37], with the large-sample multiplier. The emulation's unassisted labels come from the earlier session, a best case for a real audit. Single-epoch audits, though impractical, are reported for comparison.

2.7 How FDA-cleared OLZ devices built their references

We reviewed every 510(k) clearance by the U.S. Food and Drug Administration (FDA) under product code OLZ (automatic event detection software for polysomnography with EEG) listed by the openFDA device interface [38]: 37 clearances up to 2026-06-27. Devices cleared under other product codes, for example sleep-scoring systems cleared as polysomnographs (product code OLV), were not reviewed. For the 25 clearances since 2012, we coded four fields from each public summary: how the reference for clinical performance testing was produced, how many scorers were involved, whether the summary stated that the scorers were blinded to the device's output, and

Table 1: Primary results by task. Agreement is Cohen's kappa for sleep stages and event F1 for the event tasks.

Task	Anchoring ΔA (95% interval)	Experts > 0	Δ_{auto} range	Δ_{human} range	Session-matched inflation DiD (95% interval)
Sleep stages	0.081 (0.038 to 0.129)	11/12	+0.056 to +0.102	−0.022 to −0.011	0.091 (0.072 to 0.112)
EEG arousals	0.066 (0.011 to 0.133)	10/12	+0.018 to +0.155	−0.043 to +0.004	0.089 (0.042 to 0.145)
Respiratory events	0.129 (0.059 to 0.207)	11/12	+0.034 to +0.255	−0.132 to −0.012	0.178 (0.066 to 0.300)
Limb movements	0.103 (0.066 to 0.144)	12/12	+0.015 to +0.094	−0.016 to +0.001	0.054 (0.023 to 0.089)

Anchoring: assisted minus manual agreement of the same expert with the pre-fill, averaged over 12 experts and 5 recordings. Experts > 0: the number of the 12 experts whose mean anchoring over the task's recordings is positive. Δ_{auto}: change in the pre-fill's agreement from the manual to the assisted consensus; Δ_{human}: the same change for each expert's manual scoring against the other 11 experts' consensus (ranges over the 5 recordings). DiD $= \Delta_{\text{auto}} - \Delta_{\text{human}}$, a comparison against session-matched raters: a difference between sessions unrelated to the pre-fill lowers Δ_{human} and raises the DiD, whereas $\Delta_{\text{auto}} > 0$ in 20/20 units needs no session-matched baseline. Intervals are crossed-bootstrap intervals over experts and recordings and are descriptive. Exact sign tests over the 20 recording-task units: anchoring positive in 19/20 ($p = 4.0 \times 10^{-5}$), inflation in 20/20 ($p = 1.9 \times 10^{-6}$).

whether it said that they had worked from automatic pre-scoring by any software. The verbatim passage behind every code was recorded. A reference described only as manually scored was not counted as a statement about pre-scoring, because manual scoring can be done in software that pre-scores automatically. The AI assistant named in the statement on generative AI coded the first three fields; one of two authors checked them against each summary and recorded the fourth field, pre-scoring, during that check (Supplementary Methods). The two checkers then re-coded the three key fields of all 22 summaries by hand and, by their account, without consulting any code; each was blind for the summaries the other had checked. The corresponding author, who had directed the first coding but had not coded, settled disagreements from the summary text (Supplementary Methods). Summaries are abbreviated, so we report what they state, not what the submissions contained.

3 Results

3.1 Assisted scorings were closer to the pre-fill

Assisted scorings lay closer to the pre-fill than the same experts' manual scorings in 19 of 20 recording-task units ($p = 4.0 \times 10^{-5}$; Table 1, Figure 2a; Supplementary Table S15). Task means ranged from 0.066 to 0.129, and 10 to 12 of the 12 experts shifted toward the pre-fill in each task. The one unit without a shift, an arousal recording, was at −0.002. By the pre-specified decision rule, the result is positive, and it held under every robustness variant (Supplementary Table S4).

3.2 The references moved toward the pre-fill

The two consensuses differed on a minority of units, 3.6% of common-support staging epochs. Where they differed, the assisted consensus equaled the pre-fill in 79% (arousals) to 99% (limb movements) of the changed units, epochs for stages and 0.1-second bins for events, while the manual consensus rarely did (Figure 3c). For stages, the shares were 0.930 against 0.058 over

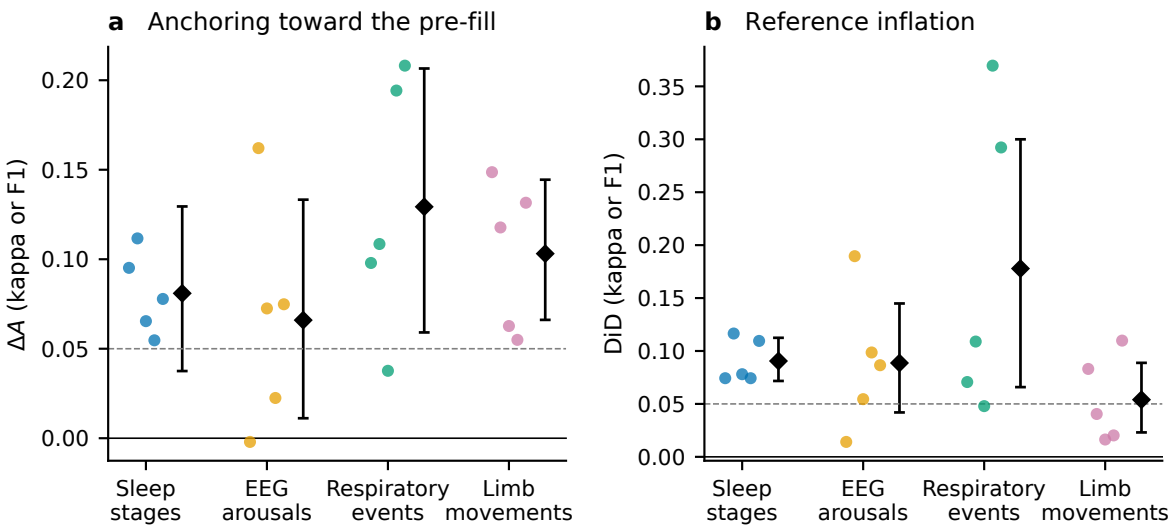


Figure 2: Primary results. (a) Anchoring: each expert's assisted minus manual agreement with the pre-fill, averaged over experts, for each recording (dots). (b) Reference inflation: the pre-fill's agreement gain from the manual to the assisted consensus, minus the gain of the same experts' manual scorings judged against the other experts' consensus. Diamonds show task means with 95% crossed-bootstrap intervals. The dashed line marks the pre-specified material effect of 0.05.

171 changed epochs. The difference was positive in 20 of 20 units ($p = 1.9 \times 10^{-6}$). Against the assisted rather than the manual consensus, the pre-fill's agreement rose in every unit ($\Delta_{\text{auto}} > 0$ in 20 of 20), a gain with no session-matched baseline. The experts' manual scorings gained at most +0.004, so the session-matched DiD was positive in 20 of 20 units ($p = 1.9 \times 10^{-6}$; Figure 2b). It was also positive in all 15 event units when consensus events shorter than the rules' minimum durations were dropped. On epochs where both consensuses exist (sleep stages), the human raters' loss shrank to −0.010 kappa (95% interval −0.024 to +0.002). On those epochs, the pre-fill's own gain stayed positive in all 5 staging recordings (mean +0.054; common-support DiD +0.064). Judged against the assisted consensus, the experts' own manual consensus (0.964 for stages) was closer than the pre-fill (0.947); the pre-fill was closer in 6 of 20 units, contrary to a pre-specified prediction (Supplementary Table S10). The assisted reference remained predominantly the experts' own judgment.

3.3 Where the pull arose

The pull concentrated where experts disagree (Figure 3a; Supplementary Tables S2 and S3). In every task the acceptance lift was near zero where all 12 experts had agreed manually and rose steadily as manual agreement fell, reaching +0.182 to +0.345 across tasks where 7 or fewer agreed. Among stages the lift was largest for N1 (+0.173; 95% interval +0.098 to +0.254), and N2 (+0.040) was the only stage whose descriptive interval included zero. For respiratory events, anchoring was larger for hypopneas (+0.234) than for apneas (+0.056).

Experts also adopted labels that most colleagues had rejected. Where at least 9 of the other 11 experts' manual labels rejected the pre-fill's, the lift was +0.169 for stages (95% interval +0.074 to +0.279) and +0.231 to +0.374 for the event tasks. It was positive in 19 of 20 units and zero in the other, a limb-movement recording ($p = 3.8 \times 10^{-6}$ over the 19 nonzero units; Figure 3b).

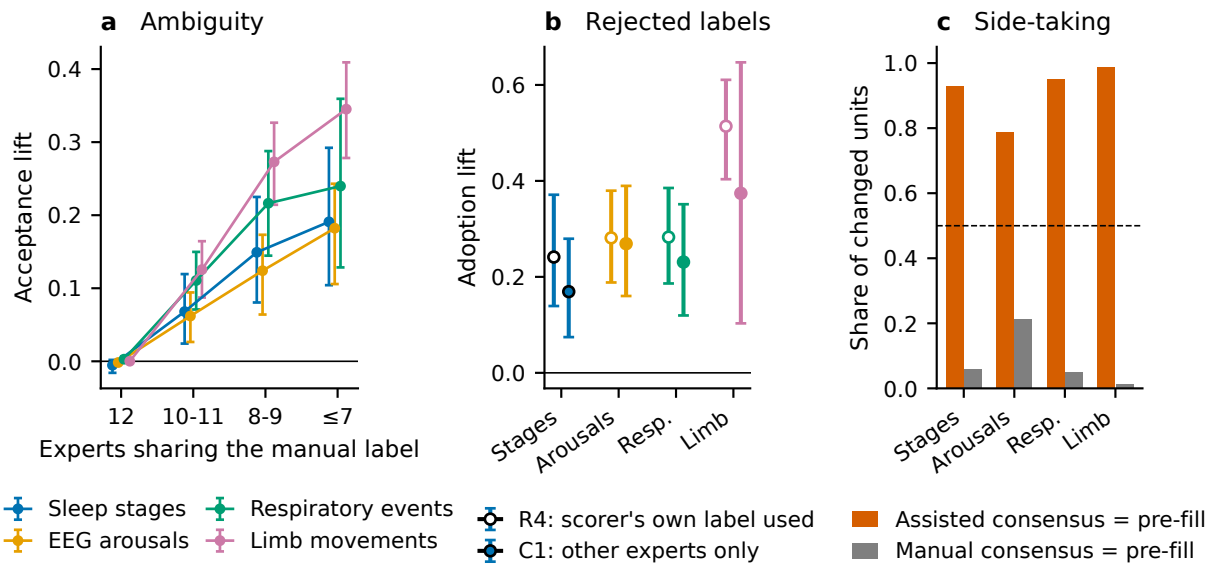


Figure 3: Where the pull arose. (a) Acceptance lift (probability that the assisted label equals the pre-fill, minus the same probability for the manual label), by the number of the 12 experts who shared the most frequent manual label. (b) The same lift on labels that most experts rejected manually, with units selected using the scored expert's own label (open) or the other 11 experts' labels only (filled; events counted per event). Error bars are 95% crossed-bootstrap intervals. (c) Side-taking: among units where the two consensuses differ, the share where each consensus equals the pre-fill. The dashed line marks equal shares.

Where the pre-fill agreed with the manual consensus, assistance raised the experts' agreement with each other. Where it disagreed, agreement fell in three of four tasks (stages: 0.593 to 0.528; Supplementary Table S13), as some experts followed the pre-fill and others did not.

3.4 Algorithms the experts never saw

YASA and GSSC disagreed with the pre-fill on 1,052 and 794 sleep-stage epochs. On these epochs the experts' assisted labels moved toward the pre-fill (+0.162 and +0.183) and away from the stagers (−0.135, 95% interval −0.206 to −0.076, and −0.156), in every recording (Figure 4a; Supplementary Table S11). Where the two consensuses differed on such epochs, the assisted consensus never took the stager's label, whereas the manual consensus took it in 93% (YASA) and 96% (GSSC) of epochs.

Against the assisted consensus, both stagers lost agreement in four of five recordings while the pre-fill gained (Table 2). The pre-fill's lead over YASA (exploratory) grew from +0.079 to +0.194 kappa, and over GSSC from +0.011 to +0.115. GSSC agreed better than the pre-fill with the manual consensus in 3 of 5 recordings and worse with the assisted consensus in all five (Figure 4b). On common support, the growth of the lead, which needs no human baseline, was positive in all five recordings for both stagers (mean +0.081 and +0.073 kappa).

3.5 Clinical indices

The two consensuses differed by 0.2 to 3.7 respiratory events, 0.1 to 3.7 arousals and 0.4 to 12.6 limb movements per hour of sleep. No recording changed its respiratory severity category

Table 2: Two independent stagers that no expert saw, judged against the manual and the assisted consensus (sleep stages, kappa).

Recording	YASA manual / assisted	GSSC manual / assisted	Pre-fill's lead over YASA manual / assisted	Pre-fill's lead over GSSC manual / assisted	Δ_{auto}, common support
SN1[a]	0.812 / 0.821	0.818 / 0.830	−0.024 / +0.029	−0.029 / +0.020	+0.043
SN2	0.709 / 0.667	0.758 / 0.701	+0.107 / +0.252	+0.058 / +0.218	+0.076
SN3	0.753 / 0.706	0.865 / 0.847	+0.101 / +0.204	−0.011 / +0.063	+0.032
SN4	0.795 / 0.739	0.841 / 0.821	+0.107 / +0.227	+0.062 / +0.145	+0.046
SN5	0.736 / 0.671	0.865 / 0.802	+0.105 / +0.259	−0.024 / +0.128	+0.074
Mean			+0.079 / +0.194	+0.011 / +0.115	+0.054

Lead: the pre-fill's kappa minus the stager's, against the same consensus; a negative lead means the stager agreed better. Δ_{auto}, common support: the pre-fill's change from the manual to the assisted consensus on epochs where both consensuses exist. [a]No C4-M1 channel; both stagers used Cz-M1.

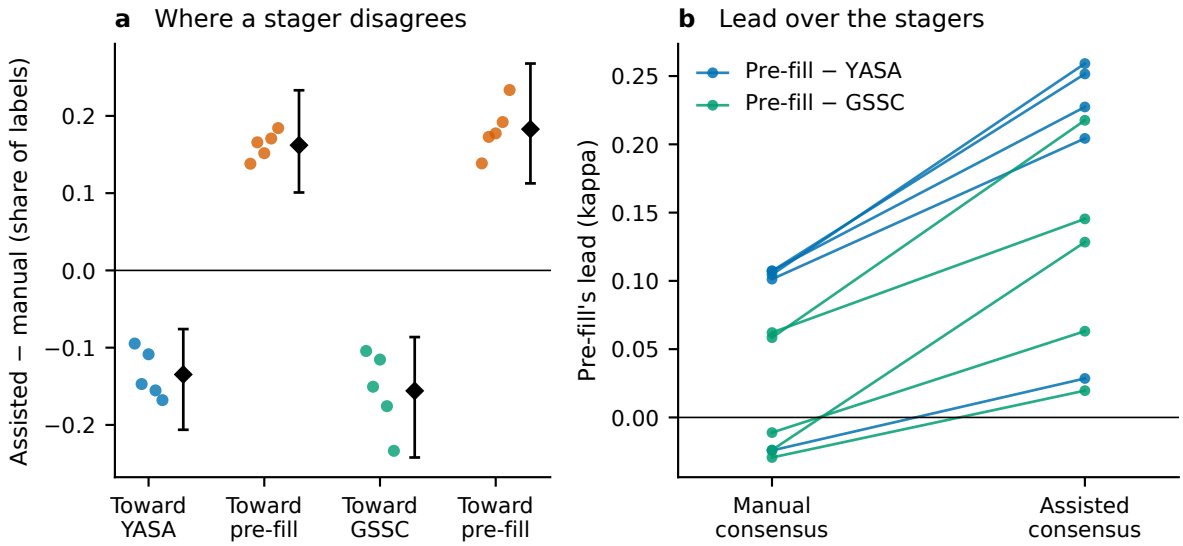


Figure 4: Two sleep stagers that no expert saw. (a) On epochs where a stager and the pre-fill disagree, each expert's assisted minus manual share of labels equal to the stager's and to the pre-fill's (dots: recordings; diamonds: means with 95% crossed-bootstrap intervals). (b) The pre-fill's lead in kappa over each stager, judged against the manual and against the assisted consensus; each line is a recording.

Table 3: Unassisted audits of 30-minute blocks, emulated on PSG-IPA (4,857 epochs in 87 blocks; percent agreement against the unassisted consensus).

Quantity	Unassisted target	Assisted reference (bias)	9-of-12 assisted (bias)	10% of blocks D1 / D2	20% of blocks D1 / D2
Pre-fill agreement	0.898	0.947 (+0.048)	0.975 (+0.077)	±0.098 / ±0.024	±0.062 / ±0.016
YASA agreement	0.850	0.839 (−0.011)	0.864 (+0.014)	±0.117 / ±0.028	±0.077 / ±0.019
Pre-fill minus YASA	0.049	0.108 (+0.059)	0.112 (+0.063)	±0.080 / ±0.045	±0.056 / ±0.030

Audits draw whole blocks at random within each recording; 10% of blocks hold 12% and 20% hold 21% of the epochs. Audit columns give 95% half-widths from exact formulas for stratified sampling of blocks, with the population variances and the large-sample multiplier, a known-variance best case (Supplementary Table S14 gives the t-multiplier values). D1: the audited blocks alone; D2: the difference estimator combining them with the assisted consensus on all epochs. Both are unbiased by construction. Biases and half-widths are computed from unrounded values. Supplementary Table S14 gives other audit sizes and the single-epoch audits.

(Supplementary Table S12, which also gives, per task, how often the pre-fill's index lay closer to the assisted than to the manual consensus).

3.6 An unassisted audit of blocks

On PSG-IPA (Table 3, Figure 5), the assisted reference overstated the pre-fill's percent agreement with the unassisted consensus by +0.048 and raised its lead over YASA from 0.049 to 0.108. The 9-of-12 assisted reference overstated both further (+0.077 and +0.063), because it keeps the epochs experts agree on, where the pre-fill is nearly always right. When whole 30-minute blocks were audited, the difference estimator's 95% half-width for the lead fell below the assisted reference's bias from 10% of blocks, 12% of epochs. There the half-width was 0.045, or 0.0586 with the t multiplier for 5 degrees of freedom, against a bias of 0.0593. Single epochs needed 5%, and the audited blocks alone needed more (Supplementary Table S14).

3.7 How FDA-cleared OLZ devices built their references

Of the 25 clearances under product code OLZ since 2012, 22 reported clinical performance data (Supplementary Table S9). Their references were experts scoring independently for the study in 10 (three scorers in 8), a consensus not described as independent in 4, routine clinical scoring in 3, a daily-practice panel in 1, another cleared algorithm's unedited output in 1, and not described in 3. No summary reported whether reference scorers had worked from automatic pre-scoring by any software, and only 1 stated that they had no access to the device's own output. The coded reference passage of 10 summaries describes the reference as manually scored, which under the coding rule is not a statement about pre-scoring. Blinding to the device was implied by the design in 5, stated without an object in 2, not applicable in 1 and not addressed in 13. In the second coding, on the summaries for which each coder was blind to the AI's checked codes, the coders agreed with those codes on 8 of 10 to 12 of 12 per field (18 to 22 of 22 overall; kappa 0.636 to 1.000; Supplementary Table S17). Adjudicating the 8 disagreements changed 1 code, SomnoMetry's reference, to not described (Supplementary Table S18).

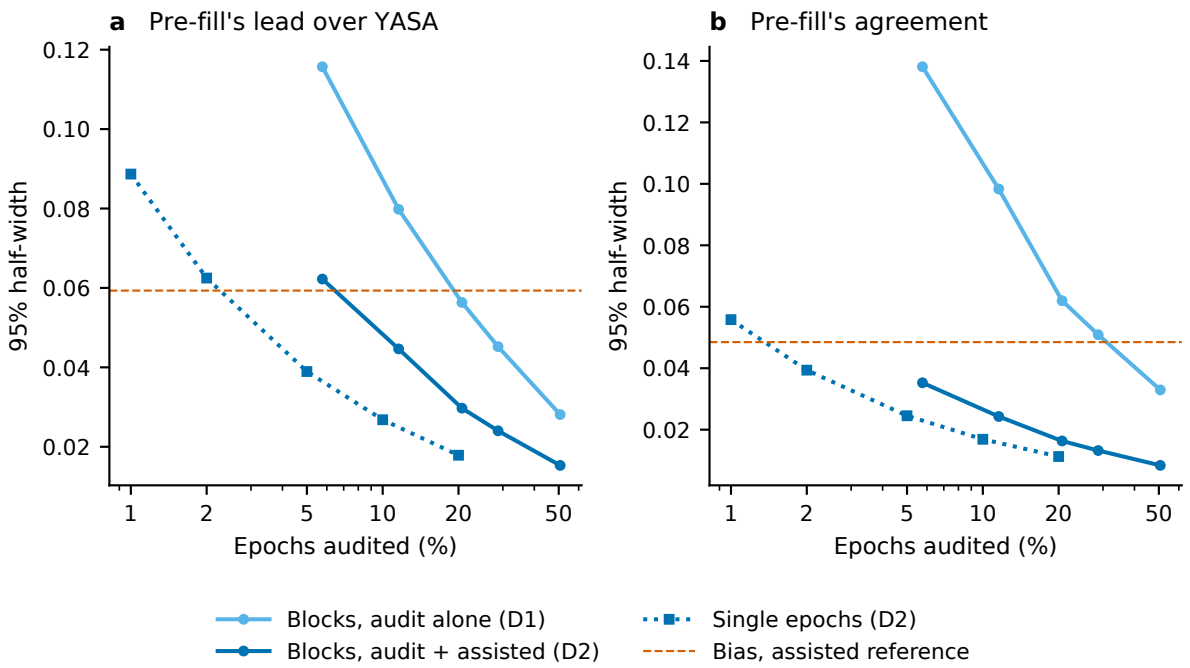


Figure 5: Unassisted audits on PSG-IPA. 95% half-widths of the audit estimates of (a) the pre-fill's lead over YASA and (b) the pre-fill's agreement with the unassisted consensus, against the share of epochs audited: random 30-minute blocks audited alone (D1) or combined with the assisted consensus by the difference estimator (D2), and single epochs (D2). The dashed line is the bias of the assisted reference.

4 Discussion

In this dataset, references built by editing an algorithm's pre-scoring were closer to it than the same experts' unassisted references, in all four tasks and most where experts disagreed. The shift went toward the algorithm the experts saw and away from two they did not. Against the assisted reference, the pre-fill outscored a stager that had matched it against the experts' unassisted judgment. In a benchmark of a commercial autoscorer, a thorough review changed 11.6% of the staging epochs and introduced a bias toward one expert scorer [39].

What the data establish. Our design cannot settle whether assistance made the references more or less accurate. Against the standard we used, the same experts' unassisted judgment, an assisted reference lies closer to the assisting algorithm, and two checks narrow the alternatives. First, the shift was consistent with one specific to the algorithm shown. Where two stagers that the experts never saw disagreed with the pre-fill, the experts moved toward the pre-fill and away from both, so a general drift between sessions toward more typical or algorithm-like scoring does not explain it. A drift toward the conventions of this center, on whose scorings the pre-fill's stager was developed and the other two were not, would produce the same pattern and cannot be separated from anchoring in this design. Second, experts adopted labels that most colleagues had rejected, with the units selected without the experts' own judgments. Neither observation shows that the adopted labels were wrong. Many scoring decisions apply thresholds by eye, such as a limb movement's amplitude [29] or a hypopnea's desaturation, and experts may share misses that a detector catches. The largest adoption lift, for limb movements, and the larger anchoring for hypopneas than for apneas are compatible with such corrections. N1 has no threshold of the kind

N3 and limb movements have, and it showed the largest stage lift; there, anchoring is the likelier reading. Keeping a pre-filled label costs nothing and changing it costs an action [16]. "Anchoring" here names the outcome, not a process. For the qualitative claim the distinction does not matter: a reference edited from one algorithm's output copies its errors with its correct labels and cannot arbitrate between that algorithm and its competitors.

Consequences for validation. In our data the assisted reference overstated the pre-fill's percent agreement by +0.048, as much as its entire unassisted gap to YASA (0.049). In kappa, it turned GSSC from level with the pre-fill into trailing it in every recording. Anchoring is a shared influence rather than independent noise, so adding experts does not remove it: in both model configurations the bias in the pre-fill's favor grew with the size of the assisted panel (Supplementary Methods and Figure S1). A stricter consensus does not help either: it keeps the easy epochs and drops those where algorithms differ, and in our data it widened the pre-fill's lead. Clinical indices changed little and no severity category changed, so the clearest consequence concerns the comparison and clearance of algorithms rather than diagnosis.

How often could this reach practice? In the OLZ clearances (other codes were not reviewed), the most common reference was experts scoring independently for the study (10 of 22), a design that can exclude pre-scoring. In 7 it was routine clinical scoring, a daily-practice panel or not described, and no summary says whether the scorers worked from pre-scoring. Where laboratory software pre-scores automatically, routine scoring may be assisted [13].

Recommendations. First, validation studies should report whether reference scorers saw any algorithm's output, and which. STARD 2015 item 13b and STARD-AI [40, 41], the QUADAS-2 reference-standard domain [42] and TRIPOD+AI [43] already ask whether the reference was assessed without knowledge of the index test; a reference edited from the index algorithm's output answers in the negative by design. The FDA's 2025 draft guidance on AI-enabled devices asks how device outputs were blinded from the reference-standard determination [44]. Two items are new: pre-scoring by any software, and a reference built with one algorithm's help and used to judge another, the comparison that in our data favored the assisting algorithm. The wording matters: "computer-assisted scoring" has meant both manual scoring in software [45, 46] and editing an automatic scoring [11]. Second, where assisted references are used, a random sample of 30-minute blocks, drawn and concealed in advance, should be scored without assistance by scorers who have not seen the algorithm's labels for them. This unassisted pass comes before the assisted pass, or is done by a separate panel that has not worked with the algorithm. Every epoch of an audited block is re-scored, so the software must withhold its pre-scoring there. In our data, auditing 10% of blocks and combining the audit with the assisted reference [36] estimated the algorithm's lead within 0.045, a known-variance best case, or 0.059 with the audit's own degrees of freedom, about the bias it corrects. The fraction is this dataset's grid value; an audit elsewhere is sized so that its half-width falls below the smallest difference that matters. A real audit by experts who also work with the algorithm would be less independent. Asking each expert for a committed judgment before revealing the suggestion is an alternative. Anchoring varied widely between experts (Supplementary Table S16), as automation-bias research would predict [47]. Third, references produced with one algorithm's assistance should not be used to compare it with others. Datasets should document any pre-annotation and the model behind it [48], since reusing such labels to train new models passes that algorithm's decisions on.

Limitations. The data come from one center, with 12 experts, five recordings per task and one family of research pre-fill algorithms by the dataset's authors. The pre-fill's stager was developed on this center's scorings and YASA and GSSC on cohorts elsewhere, so part of the pre-fill's lead may be real. Commercial autoscoring, other laboratories and scorers trained elsewhere may anchor more or less. The recordings sit at five percentiles of automatic-versus-clinical agreement (12.5th to 87.5th), so only the two tails, including the recordings where the algorithm fails most, are unrepresented. The assisted session always came at least four months after the manual one. Whether these experts used the algorithms in routine work before or between the sessions is not documented. At center level, the arousal detector's reviewed output had produced part of the clinical arousal references before this dataset was scored (220 of 472 routine recordings [13]), and the staging pre-fill comes from a stager developed on the center's clinical scorings [31], which version is not documented. Exposure before the manual session makes the measured shift a lower bound; exposure between the sessions would also move unassisted judgment toward the algorithm and inflate it. The negative control is consistent with a shift specific to the pre-fill rather than toward algorithm-like scoring in general, but cannot detect one toward conventions that only the pre-fill encodes. The selection of rejected labels assumes that experts label independently within a session. The clinical indices pool apneas and hypopneas without a documented hypopnea rule. The plans were dated and hashed but not registered, and the protocol and first amendment were committed with their results. The second coders and the adjudicator were authors; each coder was blind to the first-round codes for about half of the summaries, and none was blind to the hypothesis. A prospective study with randomized session order, or a replication with planted errors [20], would address the open questions.

5 Conclusions

In this dataset, references built by editing an algorithm's pre-scoring were closer to that algorithm than the same experts' unassisted scoring, and favored it over sleep stagers the experts never saw. Validations should report how their references were made, not compare an algorithm against references built with its help, and audit a random sample of blocks scored without assistance before any exposure to the algorithm's labels for them.

References

[1] Samaneh Nasiri, Wolfgang Ganglberger, Thijs Nassi, Erik-Jan Meulenbrugge, Valdery Moura Junior, Manohar Ghanta, Aditya Gupta, Katie L Stone, Magnus Ruud Kjaer, Oliver Sum-Ping, Emmanuel Mignot, Dennis Hwang, Lynn Marie Trotti, Gari D Clifford, Umakanth Katwa, Haoqi Sun, Robert J Thomas, and M Brandon Westover. CAISR: achieving human-level performance in automated sleep analysis across all clinical sleep metrics. *SLEEP*, 48(8), 2025. ISSN 1550-9109. doi: 10.1093/sleep/zsaf134. URL `http://dx.doi.org/10.1093/sleep/zsaf134`.

[2] Antoine Guillot, Fabien Sauvet, Emmanuel H. During, and Valentin Thorey. Dreem Open Datasets: Multi-Scored Sleep Datasets to Compare Human and Automated Sleep Staging.

IEEE Transactions on Neural Systems and Rehabilitation Engineering, 28(9):1955–1965, 2020. ISSN 1558-0210. doi: 10.1109/tnsre.2020.3011181. URL http://dx.doi.org/10.1109/TNSRE.2020.3011181.

[3] Richard S. Rosenberg and Steven Van Hout. The American Academy of Sleep Medicine Inter-scorer Reliability Program: Sleep Stage Scoring. *Journal of Clinical Sleep Medicine*, 9(1):81–87, 2013. ISSN 1550-9397. doi: 10.5664/jcsm.2350. URL http://dx.doi.org/10.5664/jcsm.2350.

[4] Richard S. Rosenberg and Steven Van Hout. The American Academy of Sleep Medicine Inter-scorer Reliability Program: Respiratory Events. *Journal of Clinical Sleep Medicine*, 10 (4):447–454, 2014. ISSN 1550-9397. doi: 10.5664/jcsm.3630. URL http://dx.doi.org/10.5664/jcsm.3630.

[5] Heidi Danker-Hopfe, Peter Anderer, Josef Zeitlhofer, Marion Boeck, Hans Dorn, Georg Gruber, Esther Heller, Erna Loretz, Doris Moser, Silvia Parapatics, Bernd Saletu, Andrea Schmidt, and Georg Dorffner. Interrater reliability for sleep scoring according to the Rechtschaffen & Kales and the new AASM standard. *Journal of Sleep Research*, 18(1):74–84, 2009. ISSN 1365-2869. doi: 10.1111/j.1365-2869.2008.00700.x. URL http://dx.doi.org/10.1111/j.1365-2869.2008.00700.x.

[6] Yun Ji Lee, Jae Yong Lee, Jae Hoon Cho, and Ji Ho Choi. Interrater reliability of sleep stage scoring: a meta-analysis. *Journal of Clinical Sleep Medicine*, 18(1):193–202, 2022. ISSN 1550-9397. doi: 10.5664/jcsm.9538. URL http://dx.doi.org/10.5664/jcsm.9538.

[7] Minna Pitkänen, Henna Pitkänen, Rajdeep Kumar Nath, Sami Nikkonen, Samu Kainulainen, Henri Korkalainen, Kristín Anna Ólafsdóttir, Erna Sif Arnardottir, Sigridur Sigurdardottir, Thomas Penzel, Francesco Fanfulla, Ulla Anttalainen, Tarja Saaresranta, Ludger Grote, Jan Hedner, Richard Staats, Juha Töyräs, and Timo Leppänen. Temporal and sleep stage-dependent agreement in manual scoring of respiratory events. *Journal of Sleep Research*, 34 (3), 2025. ISSN 1365-2869. doi: 10.1111/jsr.14391. URL http://dx.doi.org/10.1111/jsr.14391.

[8] Jessie P Bakker, Marco Ross, Andreas Cerny, Ray Vasko, Edmund Shaw, Samuel Kuna, Ulysses J Magalang, Naresh M Punjabi, and Peter Anderer. Scoring sleep with artificial intelligence enables quantification of sleep stage ambiguity: hypnodensity based on multiple expert scorers and auto-scoring. *Sleep*, 46(2), 2023. ISSN 1550-9109. doi: 10.1093/sleep/zsac154. URL http://dx.doi.org/10.1093/sleep/zsac154.

[9] U.S. Food and Drug Administration. 510(k) summary K162627: EnsoSleep (EnsoData, Inc.). Premarket notification database, 2017. URL https://www.accessdata.fda.gov/cdrh_docs/pdf16/K162627.pdf.

[10] Magdy Younes, Wayne Thompson, Colleen Leslie, Tanya Egan, and Eleni Giannouli. Utility of Technologist Editing of Polysomnography Scoring Performed by a Validated Automatic

System. *Annals of the American Thoracic Society*, 12(8):1206–1218, 2015. ISSN 2325-6621. doi: 10.1513/annalsats.201411-512oc. URL http://dx.doi.org/10.1513/AnnalsATS.201411-512OC.

[11] Naresh M. Punjabi, Naima Shifa, Georg Dorffner, Susheel Patil, Grace Pien, and Rashmi N. Aurora. Computer-Assisted Automated Scoring of Polysomnograms Using the Somnolyzer System. *Sleep*, 38(10):1555–1566, 2015. ISSN 1550-9109. doi: 10.5665/sleep.5046. URL http://dx.doi.org/10.5665/sleep.5046.

[12] Matias Rusanen, Gabriel Jouan, Riku Huttunen, Sami Nikkonen, Sigríður Sigurðardóttir, Juha Töyräs, Brett Duce, Sami Myllymaa, Erna Sif Arnardottir, Timo Leppänen, Anna Sigridur Islind, Samu Kainulainen, and Henri Korkalainen. Retrospective validation of automatic sleep analysis with grey areas model for human-in-the-loop scoring approach. *Journal of Sleep Research*, 34(3), 2025. ISSN 1365-2869. doi: 10.1111/jsr.14362. URL http://dx.doi.org/10.1111/jsr.14362.

[13] Diego Alvarez-Estevez and Isaac Fernández-Varela. Large-scale validation of an automatic EEG arousal detection algorithm using different heterogeneous databases. *Sleep Medicine*, 57:6–14, 2019. ISSN 1389-9457. doi: 10.1016/j.sleep.2019.01.025. URL http://dx.doi.org/10.1016/j.sleep.2019.01.025.

[14] Peter Anderer, Arnaud Moreau, Michael Woertz, Marco Ross, Georg Gruber, Silvia Parapatics, Erna Loretz, Esther Heller, Andrea Schmidt, Marion Boeck, Doris Moser, Gerhard Kloesch, Bernd Saletu, Gerda M. Saletu-Zyhlarz, Heidi Danker-Hopfe, Josef Zeitlhofer, and Georg Dorffner. Computer-Assisted Sleep Classification according to the Standard of the American Academy of Sleep Medicine: Validation Study of the AASM Version of the Somnolyzer 24×7. *Neuropsychobiology*, 62(4):250–264, 2010. ISSN 1423-0224. doi: 10.1159/000320864. URL http://dx.doi.org/10.1159/000320864.

[15] Magdy Younes, Jill Raneri, and Patrick Hanly. Staging Sleep in Polysomnograms: Analysis of Inter-Scorer Variability. *Journal of Clinical Sleep Medicine*, 12(6):885–894, 2016. ISSN 1550-9397. doi: 10.5664/jcsm.5894. URL http://dx.doi.org/10.5664/jcsm.5894.

[16] Diego Alvarez-Estevez and Roselyne M. Rijsman. Computer-assisted analysis of polysomnographic recordings improves inter-scorer associated agreement and scoring times. *PLOS ONE*, 17(9):e0275530, 2022. ISSN 1932-6203. doi: 10.1371/journal.pone.0275530. URL http://dx.doi.org/10.1371/journal.pone.0275530.

[17] Amos Tversky and Daniel Kahneman. Judgment under Uncertainty: Heuristics and Biases. *Science*, 185(4157):1124–1131, 1974. ISSN 1095-9203. doi: 10.1126/science.185.4157.1124. URL http://dx.doi.org/10.1126/science.185.4157.1124.

[18] Kate Goddard, Abdul Roudsari, and Jeremy C Wyatt. Automation bias: a systematic review of frequency, effect mediators, and mitigators. *Journal of the American Medical Informatics Association*, 19(1):121–127, 2012. ISSN 1527-974X. doi: 10.1136/amiajnl-2011-000089. URL http://dx.doi.org/10.1136/amiajnl-2011-000089.

[19] Thomas Dratsch, Xue Chen, Mohammad Rezazade Mehrizi, Roman Kloeckner, Aline Mähringer-Kunz, Michael Püsken, Bettina Baeßler, Stephanie Sauer, David Maintz, and Daniel Pinto dos Santos. Automation Bias in Mammography: The Impact of Artificial Intelligence BI-RADS Suggestions on Reader Performance. *Radiology*, 307(4), 2023. ISSN 1527-1315. doi: 10.1148/radiol.222176. URL http://dx.doi.org/10.1148/radiol.222176.

[20] Benedikt Holm, Arnar Óskarsson, Björn Elvar Þorleifsson, Hörður Þór Hafsteinsson, Sigríður Sigurðardóttir, Heiður Grétarsdóttir, Kenan Hoelke, Gabriel Marc Marie Jouan, Thomas Penzel, Erna Sif Arnardottir, and María Óskarsdóttir. World of ScoreCraft: Novel Multi-Scorer Experiment on the Impact of a Decision Support System in Sleep Staging. *Journal of Sleep Research*, 35(1), 2026. ISSN 1365-2869. doi: 10.1111/jsr.70113. URL http://dx.doi.org/10.1111/jsr.70113.

[21] Yevgeni Berzak, Yan Huang, Andrei Barbu, Anna Korhonen, and Boris Katz. Anchoring and Agreement in Syntactic Annotations. In *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing*, pages 2215–2224. Association for Computational Linguistics, 2016. doi: 10.18653/v1/d16-1239. URL http://dx.doi.org/10.18653/v1/d16-1239.

[22] Hope Schroeder, Deb Roy, and Jad Kabbara. Just Put a Human in the Loop? Investigating LLM-Assisted Annotation for Subjective Tasks. arXiv preprint arXiv:2507.15821, 2025. URL https://arxiv.org/abs/2507.15821.

[23] Penny Whiting, Anne W.S. Rutjes, Johannes B. Reitsma, Afina S. Glas, Patrick M.M. Bossuyt, and Jos Kleijnen. Sources of Variation and Bias in Studies of Diagnostic Accuracy: A Systematic Review. *Annals of Internal Medicine*, 140(3):189–202, 2004. ISSN 1539-3704. doi: 10.7326/0003-4819-140-3-200402030-00010. URL http://dx.doi.org/10.7326/0003-4819-140-3-200402030-00010.

[24] Andrew Worster and Christopher Carpenter. Incorporation bias in studies of diagnostic tests: how to avoid being biased about bias. *CJEM*, 10(02):174–175, 2008. ISSN 1481-8043. doi: 10.1017/s1481803500009891. URL http://dx.doi.org/10.1017/s1481803500009891.

[25] Sem L. Kampman, Rosalie L. Neijzen, Kees P. J. Braun, Sandra M. A. van der Salm, and Willem M. Otte. Incorporation bias in clinical prediction models for epilepsy. *Epilepsia Open*, 11(3):1061–1062, 2026. ISSN 2470-9239. doi: 10.1002/epi4.70252. URL http://dx.doi.org/10.1002/epi4.70252.

[26] Diego Alvarez-Estevez. PSG-IPA: A PolySomnoGraphic Inter-scorer Performance Assessment database. PhysioNet, 2026. URL https://physionet.org/content/psg-ipa/1.0.0/. Version 1.0.0. doi:10.13026/esx0-nw71.

[27] Tom Pollard, Benjamin E. Moody, Li-wei H. Lehman, Brian J. Gow, Chrystinne Fernandes, Chen Xie, Alistair Johnson, Roger G. Mark, and Thomas Heldt. PhysioNet as a global platform for biomedical research. *Nature Health*, 1(8):792–795, 2026. ISSN 3005-0693. doi: 10.1038/s44360-026-00096-z. URL http://dx.doi.org/10.1038/s44360-026-00096-z.

[28] Ary L. Goldberger, Luis A. N. Amaral, Leon Glass, Jeffrey M. Hausdorff, Plamen Ch. Ivanov, Roger G. Mark, Joseph E. Mietus, George B. Moody, Chung-Kang Peng, and H. Eugene Stanley. PhysioBank, PhysioToolkit, and PhysioNet: Components of a New Research Resource for Complex Physiologic Signals. *Circulation*, 101(23), 2000. ISSN 1524-4539. doi: 10.1161/01.cir.101.23.e215. URL http://dx.doi.org/10.1161/01.CIR.101.23.e215.

[29] R. Ferri, S. Fulda, R.P. Allen, M. Zucconi, O. Bruni, S. Chokroverty, L. Ferini-Strambi, B. Frauscher, D. Garcia-Borreguero, M. Hirshkowitz, B. Högl, Y. Inoue, A. Jahangir, M. Manconi, C.L. Marcus, D.L. Picchietti, G. Plazzi, J.W. Winkelman, and R.S. Zak. World Association of Sleep Medicine (WASM) 2016 standards for recording and scoring leg movements in polysomnograms developed by a joint task force from the International and the European Restless Legs Syndrome Study Groups (IRLSSG and EURLSSG). *Sleep Medicine*, 26:86–95, 2016. ISSN 1389-9457. doi: 10.1016/j.sleep.2016.10.010. URL http://dx.doi.org/10.1016/j.sleep.2016.10.010.

[30] Bob Kemp and Marco Roessen. Polyman: a free(ing) viewer for standard EDF(+) recordings and scorings. In *Sleep-Wake Research in The Netherlands*. Dutch Society for Sleep-Wake Research, 2007.

[31] Diego Alvarez-Estevez and Roselyne M. Rijsman. Inter-database validation of a deep learning approach for automatic sleep scoring. *PLOS ONE*, 16(8):e0256111, 2021. ISSN 1932-6203. doi: 10.1371/journal.pone.0256111. URL http://dx.doi.org/10.1371/journal.pone.0256111.

[32] Jacob Cohen. A Coefficient of Agreement for Nominal Scales. *Educational and Psychological Measurement*, 20(1):37–46, 1960. ISSN 1552-3888. doi: 10.1177/001316446002000104. URL http://dx.doi.org/10.1177/001316446002000104.

[33] Art B. Owen. The pigeonhole bootstrap. *The Annals of Applied Statistics*, 1(2), 2007. ISSN 1932-6157. doi: 10.1214/07-aoas122. URL http://dx.doi.org/10.1214/07-AOAS122.

[34] Raphael Vallat and Matthew P Walker. An open-source, high-performance tool for automated sleep staging. *eLife*, 10, 2021. ISSN 2050-084X. doi: 10.7554/elife.70092. URL http://dx.doi.org/10.7554/eLife.70092.

[35] Jevri Hanna and Agnes Flöel. An accessible and versatile deep learning-based sleep stage classifier. *Frontiers in Neuroinformatics*, 17:1086634, 2023. ISSN 1662-5196. doi: 10.3389/fninf.2023.1086634. URL http://dx.doi.org/10.3389/fninf.2023.1086634.

[36] Anastasios N. Angelopoulos, Stephen Bates, Clara Fannjiang, Michael I. Jordan, and Tijana Zrnic. Prediction-powered inference. *Science*, 382(6671):669–674, 2023. ISSN 1095-9203. doi: 10.1126/science.adi6000. URL http://dx.doi.org/10.1126/science.adi6000.

[37] William G. Cochran. *Sampling Techniques*. Wiley, New York, 3rd edition, 1977.

[38] U.S. Food and Drug Administration. 510(k) premarket notifications, product code OLZ. openFDA device 510(k) interface, https://api.fda.gov/device/510k.json; public summaries at https://www.accessdata.fda.gov, 2026. Accessed 29 September 2026.

[39] Bryan Peide Choo, Yingjuan Mok, Hong Choon Oh, Amiya Patanaik, Kishan Kishan, Animesh Awasthi, Siddharth Biju, Soumya Bhattacharjee, Yvonne Poh, and Hang Siang Wong. Benchmarking performance of an automatic polysomnography scoring system in a population with suspected sleep disorders. *Frontiers in Neurology*, 14:1123935, 2023. ISSN 1664-2295. doi: 10.3389/fneur.2023.1123935. URL http://dx.doi.org/10.3389/fneur.2023.1123935.

[40] Patrick M Bossuyt, Johannes B Reitsma, David E Bruns, Constantine A Gatsonis, Paul P Glasziou, Les Irwig, Jeroen G Lijmer, David Moher, Drummond Rennie, Henrica C W de Vet, Herbert Y Kressel, Nader Rifai, Robert M Golub, Douglas G Altman, Lotty Hooft, Daniël A Korevaar, and Jérémie F Cohen. STARD 2015: an updated list of essential items for reporting diagnostic accuracy studies. *BMJ*, page h5527, 2015. ISSN 1756-1833. doi: 10.1136/bmj.h5527. URL http://dx.doi.org/10.1136/bmj.h5527.

[41] Viknesh Sounderajah, Ahmad Guni, Xiaoxuan Liu, Gary S. Collins, Alan Karthikesalingam, Sheraz R. Markar, Robert M. Golub, Alastair K. Denniston, Shravya Shetty, David Moher, Patrick M. Bossuyt, Ara Darzi, Hutan Ashrafian, and STARD-AI Steering Committee. The STARD-AI reporting guideline for diagnostic accuracy studies using artificial intelligence. *Nature Medicine*, 31(10):3283–3289, 2025. ISSN 1546-170X. doi: 10.1038/s41591-025-03953-8. URL http://dx.doi.org/10.1038/s41591-025-03953-8.

[42] Penny F. Whiting, Anne W.S. Rutjes, Marie E. Westwood, Susan Mallett, Jonathan J. Deeks, Johannes B. Reitsma, Mariska M.G. Leeflang, Jonathan A.C. Sterne, Patrick M.M. Bossuyt, and QUADAS-2 Group. QUADAS-2: A Revised Tool for the Quality Assessment of Diagnostic Accuracy Studies. *Annals of Internal Medicine*, 155(8):529–536, 2011. ISSN 1539-3704. doi: 10.7326/0003-4819-155-8-201110180-00009. URL http://dx.doi.org/10.7326/0003-4819-155-8-201110180-00009.

[43] Gary S Collins, Karel G M Moons, Paula Dhiman, Richard D Riley, Andrew L Beam, Ben Van Calster, Marzyeh Ghassemi, Xiaoxuan Liu, Johannes B Reitsma, Maarten van Smeden, Anne-Laure Boulesteix, Jennifer Catherine Camaradou, Leo Anthony Celi, Spiros Denaxas, Alastair K Denniston, Ben Glocker, Robert M Golub, Hugh Harvey, Georg Heinze, Michael M Hoffman, André Pascal Kengne, Emily Lam, Naomi Lee, Elizabeth W Loder, Lena Maier-Hein, Bilal A Mateen, Melissa D McCradden, Lauren Oakden-Rayner, Johan Ordish, Richard Parnell, Sherri Rose, Karandeep Singh, Laure Wynants, and Patricia Logullo. TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods. *BMJ*, 385:e078378, 2024. ISSN 1756-1833. doi: 10.1136/bmj-2023-078378. URL http://dx.doi.org/10.1136/bmj-2023-078378.

[44] U.S. Food and Drug Administration. Artificial Intelligence-Enabled Device Software Functions: Lifecycle Management and Marketing Submission Recommendations. Draft Guidance for Industry and Food and Drug Administration Staff, January 2025. URL https://www.fda.gov/media/184856/download. Draft guidance issued on 7 January 2025.

[45] Samuel T. Kuna, Ruth Benca, Clete A. Kushida, James Walsh, Magdy Younes, Bethany Staley, Alexandra Hanlon, Allan I. Pack, Grace W. Pien, and Atul Malhotra. Agreement in Computer-

Assisted Manual Scoring of Polysomnograms across Sleep Centers. *Sleep*, 36(4):583–589, 2013. ISSN 1550-9109. doi: 10.5665/sleep.2550. URL http://dx.doi.org/10.5665/sleep.2550.

[46] Atul Malhotra, Magdy Younes, Samuel T. Kuna, Ruth Benca, Clete A. Kushida, James Walsh, Alexandra Hanlon, Bethany Staley, Allan I. Pack, and Grace W. Pien. Performance of an Automated Polysomnography Scoring System Versus Computer-Assisted Manual Scoring. *Sleep*, 36(4):573–582, 2013. ISSN 1550-9109. doi: 10.5665/sleep.2548. URL http://dx.doi.org/10.5665/sleep.2548.

[47] Raja Parasuraman and Dietrich H. Manzey. Complacency and Bias in Human Use of Automation: An Attentional Integration. *Human Factors*, 52(3):381–410, 2010. ISSN 1547-8181. doi: 10.1177/0018720810376055. URL http://dx.doi.org/10.1177/0018720810376055.

[48] Timnit Gebru, Jamie Morgenstern, Briana Vecchione, Jennifer Wortman Vaughan, Hanna Wallach, Hal Daumé III, and Kate Crawford. Datasheets for datasets. *Communications of the ACM*, 64(12):86–92, 2021. ISSN 1557-7317. doi: 10.1145/3458723. URL http://dx.doi.org/10.1145/3458723.